Yuhai Wang

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Education

Northeastern University — Boston, MA
Ph.D. in Civil and Environmental Engineering; — *Sep.2025 – Present*
Advisor: Gilbert Yang Ye

University of Southern California — Los Angeles, CA
M.S. in Data Analytics; — *Jan. 2023 – Jan. 2025*
Advisor: Daniel Seita

Tiangong University — Tianjin, China
B.E. in Computer Engineering; — *Aug. 2018 – May. 2022*
Advisor: Xuan Xiao

Professional Summary

Robotics researcher focused on the intersection of robot learning and representation learning to enable autonomous perception, navigation, and manipulation. Developed hierarchical frameworks and predictive world models that improve structural state representations for complex manipulation tasks. Background includes establishing reinforcement learning methods for dexterous, multi-fingered manipulation in cluttered, unmodeled environments. Proven experience in sim-to-real transfer, with validation across simulation (Isaac Lab/Gym, MuJoCo) and physical platforms (Franka, Allegro, ARX5, Go2).

Selected Publications

1. **Wang, Yuhai**; Jiawei Xia, Rongxuan Zhou, Xiao Hu, Yongliang Shi, Jing Du, Yang Ye. PRISM: PRior-guided Imagination Sampling in World Models. *CoRL 2026, Under Review.* [pdf], [project page], [code].
2. Xianyao Li, **Wang, Yuhai**; Xiao Hu, Kaleb Smith, Gilbert Yang Ye, Eric Jing Du. When to Personalize Household Object Search: A Rigidity-Gated Hybrid Policy. *International Conference on Intelligent Robots and Systems (IROS), 2026.* [pdf].
3. **Wang, Yuhai**; Xiao Hu, Yangming Shi, Yang Ye. DR-VPR: Dual-Branch Rotation-Robust Visual Place Recognition *under review, 2025.* [pdf], [project page].
4. **Wang, Yuhai**; Maryam Pishgar. Dynamic Token Selective Transformer for Aerial-Ground Person Re-Identification. *IEEE International Conference on Multimedia & Expo (ICME), 2025.* [pdf], [project page].
5. Jiang, Hao; **Wang, Yuhai***; Zhou, Hanyang*; Seita, Daniel. Learning to Singulate Objects in Packed Environments Using a Dexterous Hand. *International Symposium of Robotics Research (ISRR), 2024.* [pdf], [project page].

Research Experience

CRAFT Lab, Northeastern University — Boston, MA
Graduate Research Assistant, advised by Prof. Gilbert Yang Ye — *Sep. 2025 – present*

- Built PRISM, a prior-guided world model that plans by sampling imagined rollouts in a learned latent space (CoRL under review).
- Developed a hierarchical vision-language-action framework for humanoid navigation using terrain-aware feedback.
- Designed DR-VPR, a dual-branch method for rotation-robust visual place recognition under large viewpoint change.

- Co-developed a rigidity-gated policy that decides when to personalize household object search (IROS 2026).

SLURM Lab, University of Southern California Los Angeles, CA
Research Assistant, advised by Prof. Daniel Seita *Sep. 2023 – present*

- Built a reinforcement-learning framework for singulating objects in cluttered scenes in Isaac Gym, using a displacement-based state representation and multi-phase training; validated on Allegro and Franka hardware with RealSense sensing (ISRR 2024).
- Designed a Top-k token-selective Transformer for aerial-ground person re-identification (ICME 2025).

Institute of AI Industry Research(AIR), Tsinghua University Remote
Research Assistant, advised by Prof. Guyue Zhou & Prof. Yongliang Shi *April. 2023 – Sep. 2023*

- Built a distributed NeRF system with three-stage pose optimization (Mip-NeRF360 for pose estimation, inverted Mip-NeRF360 and dynamic low-pass filtering for robustness), and fused NeRFs across coordinate systems by estimating coarse transformations (IROS 2024).

Institute of AI Industry Research(AIR), Tsinghua University Beijing, China
Research Engineer, advised by Prof. Guyue Zhou & Prof. Xinliang Zhang *Aug. 2021 – Feb. 2022*

- Simulated and controlled the ARX5 arm in ROS/MoveIt (URDF exported from SolidWorks), with RealSense-based ArUco tracking for end-effector servoing.
- Helped build the simulation environment for the ICRA 2022 RoboMaster University Sim2Real Challenge.

Robotics Research Lab, Tiangong University Tianjin, China
Research Assistant, advised by Prof. Xuan Xiao *Oct. 2019 – Aug. 2021*

- Designed a quadruped robot with a novel four-bar-linkage leg; ran the kinematics in C, simulated gaits in Webots, and validated two gaits on hardware via MATLAB control (ICRA 2021).

Service

Reviewer

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	2026
Conference on Robot Learning (CoRL)	2026
IEEE International Conference on Multimedia & Expo (ICME)	2025
IEEE Transactions on Human-Machine Systems	2025
Agile Robotics Workshop @ ICRA	2024

Speaker

CEE Graduate Research Expo, Northeastern University	2025

SLURM Lab, USC Los Angelegs, CA
Graduate Research Assistant *Sep. 2023 – now*

WBCD Competition@ICRA 2025 Atlanta, GA
Hardware Sponsor *May. 2025*

ISE 534: Data Analytics Consulting, USC Los Angelegs, CA
Graduate Teaching Assistant *Jan. 2024 – May. 2024*

School of Computer Science and Technology, Tiangong University Tianjin, China
Academic Representative *Aug. 2018 – May. 2022*

Honors & Awards

CEE Department Fellowship	2025, 2026
First Prize, National Challenge Cup	2021
Outstanding Student Leader Award	2019, 2021
Honorable Mention, Mathematical Contest in Modeling	2020
President's Scholarship	2019, 2020
Social Activities Scholarship	2020
Off-campus Competition Scholarship	2020